

Fine-Tuning HuggingFace Embedding Models for Asymmetric Search

Train embedders on your data

Asymmetric Search: Why It Matters

- Short Queries $\neq$ Long Chunks

QUERY

What days of the week are campus tours offered at Christopher Newport University?

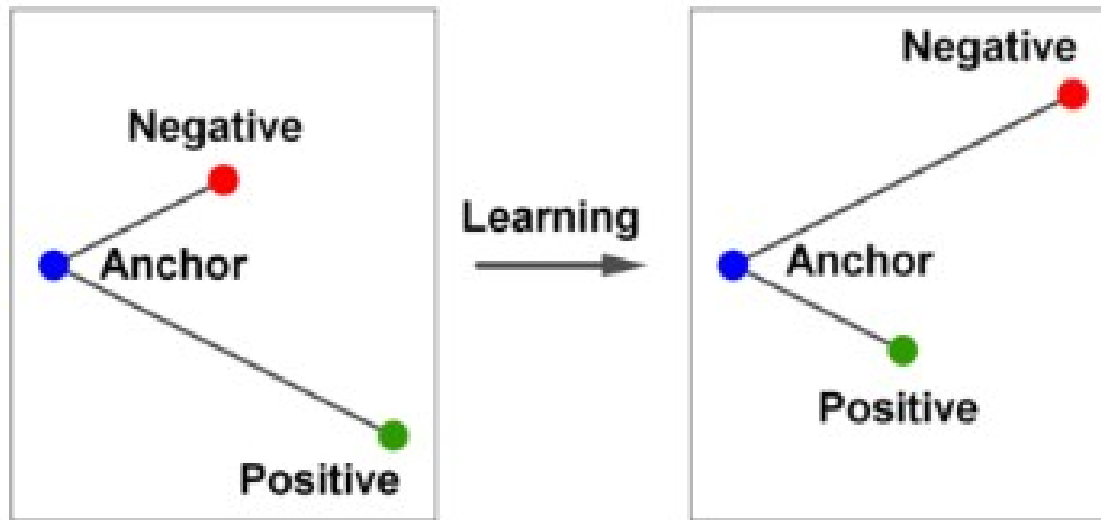
Chunk

'Come See for Yourself \n' 'Ready to picture your life as a CNU Captain? Schedule \n' 'a visit and see why so many students choose to call Chris-\n' 'topher Newport home. \n' 'Campus tours are offered Monday through Saturday \n' '(except major holidays). Visit us online at admission.cnu. \n' 'edu or call (757) 594-7015/(800) 333-4268. \n' 'We look forward to showing you why Christopher New-\n' 'port is the number one regional public university in Virginia!'

- Embedder Model must learn directional alignment
- Goal: Fine tuning improves retrieval accuracy

Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



Want to push negative away and pull positive closer

Loss Functions

- MultipleNegativesRankingLoss (MNRL)
- Uses 'in-batch' negatives
- ex. For this batch of 4

What days of the week are campus tours offered...	Come See for Yourself \nReady to picture your ...
Which minor subjects are offered under the cat...	Health Studies
Which minors require 143 credits?	Film Studies, minor
Which minor fields of study are offered in the...	Philosophy of Law, minor

Loss Functions

- MultipleNegativesRankingLoss (MNRL)
- Uses 'in-batch' negatives
- ex. For this batch of 4, **we get 3 NEGATIVES**

What days	question (or anchor)	ours offered...	Come See for You	POSITIVE	our ...
Which minor subjects are offered under the cat...			He	NEGATIVE 1	
	Which minors require 143 credits?		Film	NEGATIVE 2	
Which minor fields of study are offered in the...			Philoso	NEGATIVE 3	

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So this row will expand into 4 rows

(Anchor,POSITIVE,Negative1), (Anchor,POSITIVE,Negative2) etc

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The larger the batch the more negatives. But they are randomly picked, so easy negatives 😞

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We want hard negatives, those that are 'close' to the positive!

Why Hard Negatives?

Not all negatives are equal

Easy Negative

“How to request PTO?”

vs.

“Company stock plan details.”

Too different

Hard Negative

“How can I change my benefits?”

vs.

“What if I need to update my spouse’s benefits?”

Semantically close, teaches model what NOT to retrieve

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Hard negatives improve:

- Recall@K
- MRR
- Generalization
- Ranking quality

Loss Functions

TripletLoss

- Uses explicit (anchor, positive, **negative**)
- Teaches model to distinguish between positive passages and confusing look-alikes

Typical outcome

Big improvements in Recall@K

Sharper ranking between similar passages

Loss Functions

TripletLoss

Uses explicit (anchor, positive, negative)

How does it work?

Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



Margin is how much further out from positive that we will tolerate

Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



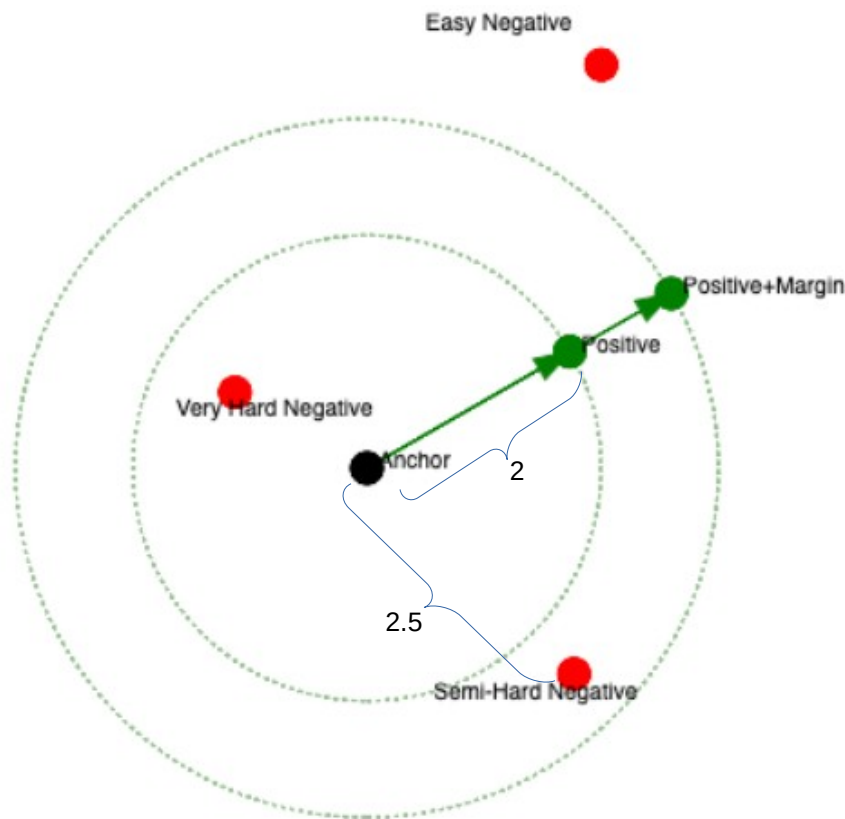
Margin=1

Loss= $\max(2-4+1, 0) \Rightarrow 0$

no loss, so nothing to learn here

Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



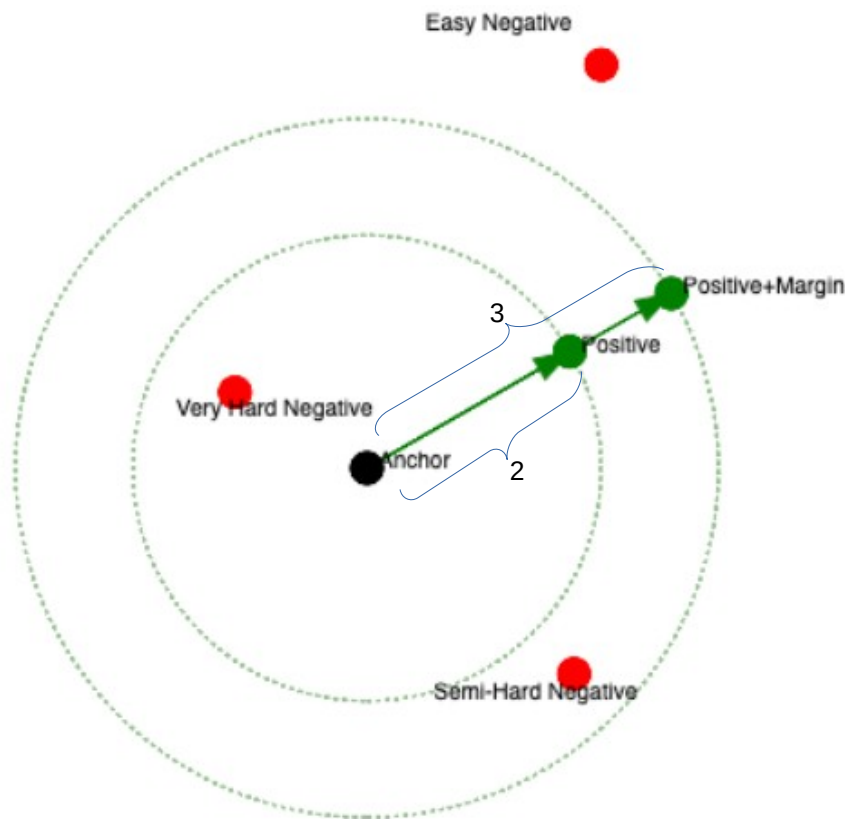
Margin=1

Loss= $\max(2-2.5+1, 0) \Rightarrow .5$

.5 loss learn from this one!

Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



Margin=1

Loss= $\max(2-3+1, 0) \Rightarrow 0$ no learning from this one

Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



- Want 'hard' negatives
- Not 'easy' negatives
- Want negatives that are 'close'

Loss Functions

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- Want 'hard' negatives
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- Want negatives that are 'close'
- **How?**

Loss Functions

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- Want 'hard' negatives
- Not 'easy' negatives
- Want negatives that are 'close'
- How?
- **Pull from Vector DB.**

Hard Negative (HN) Mining

- Encode all chunks → store in ChromaDB
- Encode question → query top-K closest chunks
- Filter true positive → keep top 3–5 as Hns

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Original Anchor	Orig Positive
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← Start with this

Hard Negative (HN) Mining

- Encode all chunks → store in ChromaDB
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- Filter true positive → keep top 3–5 as Hns

Original Anchor	Orig Positive	Hard Negative 1
Original Anchor	Orig Positive	Hard Negative 2
Original Anchor	Orig Positive	Hard Negative 3
Original Anchor	Orig Positive	Hard Negative 4

Finish
with this

Hard Negative (HN) Mining

- Encode all chunks → store in ChromaDB
- Encode question → query top-K closest chunks
- Filter true positive → keep top 3–5 as Hns
- **This dramatically expands training dataset**

Original Anchor	Orig Positive	Hard Negative 1	} New rows
Original Anchor	Orig Positive	Hard Negative 2	
Original Anchor	Orig Positive	Hard Negative 3	
Original Anchor	Orig Positive	Hard Negative 4	

New Column

Fine-Tuning

Better performance on domain-specific vocabulary (legal, medical, etc.)

- Improves retrieval precision/recall
- Reduces hallucinations in downstream RAG pipelines

2-Stage Embedder Fine-Tuning

- Stage 1: **Fine tune** model using MNRL on (question, chunk) pairs
- Embed corpus using fine tuned model
- Hard Negative Mine to expand training dataset
- Stage 2: **Fine tune** model using TripletLoss on (question, positive, hard negative)

Key Takeaways

- MNRL uses easy negatives but will probably improve model
- Triplet loss uses hard negatives and will probably dramatically improve model
- Multi-stage finetuning is generally very effective, especially for specialized domains (medicine, law, etc)